

KNN Error Rate and Bayes Error Calculation

Dataset:

Index	Color (X)	Fruit (Y)
1	Red (1)	Apple (A)
2	Red (1)	Apple (A)
3	Green (2)	Apple (A)
4	Green (2)	Banana (B)
5	Orange (3)	Banana (B)
6	Yellow (4)	Banana (B)

1. Bayes Error Calculation:

Bayes Error Formula:

$$P_{\text{error}}^{\text{Bayes}} = 1 - \sum_x P(x) * \max_y P(y|x)$$

Step 1: Priors:

- $P(\text{Red})=2/6$, $P(\text{Green})=2/6$, $P(\text{Orange})=1/6$, $P(\text{Yellow})=1/6$

Step 2: Conditionals:

- Red: $P(A|\text{Red})=1$, $P(B|\text{Red})=0$
- Green: $P(A|\text{Green})=0.5$, $P(B|\text{Green})=0.5$
- Orange: $P(B|\text{Orange})=1$
- Yellow: $P(B|\text{Yellow})=1$

Step 3: Error contributions:

- Red: 0, Green: 0.5, Orange: 0, Yellow: 0

Weighted: $(2/6)(0.5)=1/6 \approx 0.1667$

Bayes Error = 16.7%

2. KNN (k=3) Error Calculation:

For each sample, 3 nearest neighbors were chosen (excluding self). Majority voting was applied.

Summary:

- Sample 1: Pred=A, Actual=A ✓
- Sample 2: Pred=A, Actual=A ✓
- Sample 3: Pred=A, Actual=A ✓
- Sample 4: Pred=A, Actual=B ✗
- Sample 5: Pred=B, Actual=B ✓
- Sample 6: Pred=B, Actual=B ✓

Errors = $1/6 = 16.7\%$

Accuracy = $5/6 = 83.3\%$

Confusion Matrix (Actual × Predicted):

Actual \ Pred	Apple	Banana
Apple	3	0
Banana	1	2